18A_AI_ORCHESTRATION_ENGINE.md

UNPLND AI Orchestration Engine

Version 1.0

Priority: CRITICAL

Status: MVP

Purpose

The AI is not the product.

The AI is the invisible planner.

Users should never think:

"AI generated this."

Users should think:

"This evening is perfect."

The AI must disappear.

Only the experience remains.

AI Responsibilities

The AI must:

Understand the user's intent.

Plan an unforgettable experience.

Optimize travel.

Stay within budget.

Respect opening hours.

Respect weather.

Create variety.

Avoid repetition.

Think like a local.

Never think like a chatbot.

System Architecture

User Input

↓

Validation Engine

↓

Context Builder

↓

AI Request

↓

Response Validator

↓

Route Optimizer

↓

Fallback Engine

↓

Final Route

Claude Code must keep these layers separate.

Input Variables

Mood

Budget

Group Size

Transportation

Available Time

Current Time

Current Day

Location

Weather

Language

Past Plans

Favorites

Avoided Places

Accessibility Preferences

Indoor / Outdoor

Context Builder

Before calling AI always enrich the prompt with:

Nearby places

Opening hours

Travel times

Weather

Sunset time

Public holidays

Current season

User history

Premium features (if applicable)

Never send raw user input alone.

AI Personality

The AI behaves like:

A local friend.

Someone who knows the city.

Someone who genuinely wants the user to have an amazing evening.

Never behaves like ChatGPT.

Never explains reasoning unless requested.

Never mentions AI.

Planning Rules

Always start logically.

Typical structure:

Food

↓

Activity

↓

Optional Dessert

↓

Bar

↓

Late Activity

↓

End

Not every evening needs every stop.

Quality beats quantity.

Budget Rules

Never exceed the user's budget.

Target

80–95%

Leave a financial buffer.

Example

Budget

€50

Target Spend

€40–47

Never recommend impossible combinations.

Distance Rules

Walking

Ideal

<10 min

Maximum

20 min

Car

Prefer

<20 min

Public Transport

Avoid excessive transfers.

Never send users across the city unnecessarily.

Weather Rules

Rain

↓

Indoor first

Sunny

↓

Outdoor preferred

Wind

↓

Avoid rooftop suggestions

Heat

↓

Shade

Water

Indoor alternatives

Always adapt.

Time Rules

Morning

Coffee

Brunch

Walk

Afternoon

Activities

Shopping

Nature

Evening

Dinner

Bars

Night

Clubs

Late Food

Never recommend closed venues.

Variety Rules

Avoid:

Restaurant

↓

Restaurant

↓

Restaurant

Instead:

Restaurant

↓

Experience

↓

Drinks

↓

Walk

↓

Memory

The evening should tell a story.

Hidden Gem Priority

Whenever possible:

Prefer unique places over chains.

Unless the user explicitly prefers chains.

The AI should surprise positively.

Couple Mode

Increase:

Romantic atmosphere

Sunset

Views

Cozy cafés

Slow pacing

Lower noise

Friend Group Mode

Increase:

Energy

Activities

Bars

Group-friendly venues

Photo opportunities

Solo Mode

Increase:

Museums

Coffee

Walks

Books

Viewpoints

Quiet experiences

Family Mode (Future)

Safe.

Kids.

Parks.

Early closing.

Response Format

The AI must always return structured JSON.

Never Markdown.

Never plain text.

Schema

{ "summary": "", "estimated_cost": 0, "estimated_duration": "", "weather_note": "", "route": [{ "type": "", "title": "", "reason": "", "estimated_time": "", "price": "", "coordinates": "", "alternative": "" }] }

Strict schema validation.

Validation Engine

Reject responses if:

Budget exceeded.

Closed venues.

Missing coordinates.

Travel impossible.

Missing alternatives.

Invalid JSON.

Retry automatically.

Retry Strategy

Attempt 1

Normal

Attempt 2

Increase search radius

Attempt 3

Reduce constraints

Attempt 4

Offer manual editing

Never fail silently.

Hallucination Protection

Never invent places.

Never invent addresses.

Never invent opening hours.

Only recommend verified locations.

Learning

Future Version

Learn:

Favorite cuisines

Favorite budgets

Favorite moods

Planning frequency

Ignored recommendations

Never become creepy.

Never over-personalize.

Ethics

Never manipulate users.

Never prioritize sponsored locations over better recommendations.

Never discriminate.

Never recommend unsafe activities.

Never recommend illegal activities.

Performance

Context Building

<500ms

AI Response

Target

<6 seconds

Validation

<300ms

Total

<8 seconds

Failure Modes

If no valid route exists:

Suggest home experiences.

Movie night.

Board games.

Cooking together.

Sunset drive.

Indoor cafés.

Always provide value.

DO NOT

Do NOT expose prompts.

Do NOT expose reasoning.

Do NOT mention AI providers.

Do NOT output raw text.

Do NOT recommend closed places.

Do NOT exceed budget.

Do NOT generate unrealistic routes.

Golden Rule

The AI should feel like the smartest friend in the city.

Not the smartest computer.

Claude Instructions

Never bypass validation.

Never trust AI output directly.

Always verify.

Always optimize.

Always prioritize experience over novelty.

The route should feel handcrafted.

Not generated.

END OF 18A_AI_ORCHESTRATION_ENGINE.md